Spatial Context Did Not Help: A U-Net Loses to a Per-Cell Tree Model for Harmful-Algal-Bloom Forecasting, Measured Twice

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Abstract—A per-cell classifier forecasting harmful-algal-bloom risk cannot see an eddy, a front, or the shape of a bloom spanning several grid cells — the standard architectural objection to per-cell models, and the standard argument for a convolutional alternative. We test that objection directly: a single-frame U-Net given a 14-channel ocean-state image (seven physical fields, each paired with an observed-mask channel) is trained to forecast t+3 bloom probability and scored against the existing per-cell LightGBM model, reloaded and rescored on the U-Net's own 575,064 test rows so the comparison is identical-row rather than merely same-sized. The U-Net beats a persistence baseline (PR-AUC 0.5438 vs. 0.5178) but loses clearly to LightGBM (0.6680) on every one of four metrics. We re-ran the full pipeline live while writing this paper and reproduce the original measurement to three to four decimal places, and report an unrelated engineering hazard found in building the comparison: importing torch before loading a LightGBM artifact with joblib in the same process segfaults with no traceback, which is why the tabular baseline is scored in a subprocess. The negative result is evidence against reaching for a spatiotemporal architecture (ConvLSTM) next, since this agenda item was deliberately ordered to make that inference available from a single-frame test.

Index Terms—harmful algal blooms, spatial deep learning, U-Net, gradient boosting, negative results, reproducibility

I. Introduction

A per-cell classifier scores every grid cell of a harmful-algal-bloom early-warning system independently: the same gradient-boosted tree, applied row by row, with no mechanism to see an eddy, a front, or the shape of a bloom spanning several cells. The natural architectural objection is that this throws away information a spatial model could use, and the natural fix is a convolutional model that sees the whole field at once. This paper reports a controlled test of that objection for one operational system and finds it does not hold, at least not the way it is usually tried: a single-frame U-Net [1] given a 14-channel ocean-state image (seven physical fields, each paired with an observed-mask channel) loses clearly to the existing per-cell LightGBM model on 575,064 identical held-out rows, despite the two being scored through the same metrics on the same test period by construction.

The result matters beyond this one system for two reasons. First, it was tested rigorously rather than assumed: the existing tabular model was reloaded and rescored on the spatial model's own exact test rows rather than cited from a README, closing the gap between "the spatial model scored lower than a number we remembered" and "the spatial model scored lower than the same model, run again, on the same water." Second, the negative result is informative about what to try next, not just what not to do: the architecture plan that scoped this work deliberately ordered a spatial-only model before a spatial-plus-temporal one (a ConvLSTM [2]) specifically so that a negative single-frame result would argue against the more expensive next step, not merely postpone it. It does.

A. Contributions

1) A matched, identical-row comparison between a spatial U-Net and the shipped per-cell LightGBM model for t+3 harmful-algal-bloom probability, built so that a fair comparison does not depend on trusting two numbers measured at different times on different row sets.
2) A live re-verification: rather than reporting the 2026-08-31 build's numbers as a standing fact, this paper re-ran the full training-and-comparison pipeline on 2026-09-09 and confirms the original result to three to four decimal places on every metric (§IV).
3) A stated engineering hazard specific to mixing this stack: importing torch and then loading a LightGBM artifact with joblib in the same process segfaults on this platform, with no Python traceback — silent enough that the fix (score the tabular baseline in a subprocess) had to be designed as part of making the comparison trustworthy at all, not as an afterthought (§III).
4) An explicit narrowing of the next step: the result is evidence against reaching for temporal recurrence next, not merely a placeholder result to be superseded by one.

II. Related Work

a) U-Net and focal loss: The U-Net encoder-decoder with skip connections [1] is the standard architecture for dense, per-pixel prediction from a small amount of training data, which is the regime this problem sits in:

1,534 training dates, not the tens of thousands of images a segmentation benchmark assumes. Focal loss [3] was designed for exactly the class-imbalance problem a 9.7% bloom rate creates, down-weighting the loss contribution of confidently-correct negatives so a handful of hard positive cells is not swamped; this codebase's own tabular model handles the same imbalance with a class-weighting scheme, so the two models face an equivalent, not merely similar, imbalance problem.

b) Deep learning versus trees on structured environmental data: Grinsztajn et al. [4] show tree-based models remain state-of-the-art on medium-sized tabular data even accounting for training cost, and trace the gap to inductive-bias differences rather than raw capacity. That result is about the same features presented in a different representation (rows versus columns); the comparison in this paper is a different axis, holding representation architecture fixed to what each model naturally wants (a dense image for the U-Net, rows for LightGBM) and asking instead whether spatial context itself — neighbouring cells, not just more rows of the same per-cell features — is worth the switch. Both findings point the same direction for this kind of data without being the same finding.

c) ConvLSTM and the ordering of this agenda item: Convolutional LSTMs [2] extend a recurrent architecture with convolutional gates specifically to model spatiotemporal sequences such as radar nowcasting, and are the natural reach once a single frame is judged insufficient. This paper's plan ordered the spatial-only U-Net before the spatial-plus-temporal ConvLSTM for a stated reason, not simplicity alone: if a full spatial neighbourhood in a single frame does not help, stacking several such frames in time is unlikely to be the fix, and a negative result here is therefore evidence against that next step rather than merely a smaller version of it.

III. Methods

a) Task and channels: The target is bloom_t3, the codebase's one sanctioned forward-shifted label (t+3 harmful-algal-bloom onset), already fitted train-only by the existing labelling pipeline and reused unchanged — the spatial pilot introduces no second forward shift. The input is a 14-channel image per date: seven contemporaneous physical fields (chlorophyll, sea-surface temperature, mixed-layer depth, a chlorophyll gradient, an SST gradient, Okubo–Weiss vorticity, and a wind-driven upwelling index), chosen to match the architectural objection directly — a stratification proxy, frontal-strength gradients, an eddy-structure term, and the nutrient-supply driver a per-cell model cannot see the shape of — each paired with its own observed-mask channel, since the Arabian Sea grid used here is a rectangle of ocean and land and a value channel alone cannot distinguish "zero" from "not observed here."

b) Grid and architecture: The native grid is 69×41 cells at 0.25° over the Arabian Sea, zero-padded to 72×48 so three stride-2 pooling levels reach a clean bottleneck without pooling a single-digit dimension; padded cells carry a permanently-false valid mask and contribute to neither the loss nor any reported metric. The network is a three-level U-Net with 16/32/64/128 channels per level, GroupNorm rather than BatchNorm (training batches are tens of frames, not hundreds, where BatchNorm's running statistics are unreliable), and a masked focal loss ($\alpha = 0.25$, $\gamma = 2.0$) that excludes padding and any cell with no label from the gradient entirely rather than down-weighting it. The trained checkpoint holds 485,025 parameters.

c) Matched split and training: Train/validation/test splits use the identical chronological-fraction cutoffs the tabular model's own final fit uses, so the test period is the same water at the same dates by construction: 1,534 training, 329 validation, and 329 test dates. Training uses Adam (lr 10^{-3}, batch size 16) with early stopping on validation loss (patience 8 epochs); the reported run stopped after 19 epochs in 70.7 s on Apple Silicon MPS — early stopping fired well before any wall-clock or hardware limit was approached, though neither model capacity nor training schedule was swept (§VI).

d) Matched-row scoring, and a segfault it was built to avoid: The trained U-Net's dense per-date predictions are pivoted back into tidy rows at exactly the coordinates the tabular pipeline scores, using the shared fusion layer's dense-cube construction in both directions, so the row set compared is identical rather than merely similar in size. The shipped LightGBM artifact is reloaded and rescored, and persistence is recomputed, on that identical row set — deliberately in a separate process. Measured directly on this platform: importing torch anywhere earlier in a process and then loading the LightGBM pipeline with joblib segfaults (SIGSEGV, no Python traceback), and the reverse import order does not crash but is unavailable here, since the same process must load torch to run the U-Net at all. Scoring the tabular baseline in a subprocess with no torch import is therefore not an optimisation but the only way the three-way comparison could be trusted to run at all, and is stated here because a comparison pipeline that silently crashed midway would have been a second, unrelated way for this result to look wrong for reasons that had nothing to do with which model forecasts better.

IV. Results

Table I was produced by re-running hab_early_warning.src.spatial_train in full on 2026-09-09 — training a fresh U-Net from the same seed and configuration, reloading the same shipped LightGBM artifact, and rescoring both plus persistence on the 575,064 rows (positive rate 7.59%) that run's own test split produces. Every metric matches the 2026-08-31 account to three or four decimal places: PR-AUC 0.5438 (originally 0.544), ROC-AUC 0.8966 (0.897), Brier 0.0577 (0.058), TSS 0.6545 (0.655) for the spatial model,

TABLE I
All three models scored on the identical 575,064 test rows. Higher is better except Brier. Re-run live on 2026-09-09; the 2026-08-31 measurement this reproduces is quoted in the text.

Model	PR-AUC	ROC-AUC	Brier	TSS
Spatial U-Net	0.5438	0.8966	0.0577	0.6545
LightGBM (reloaded)	0.6680	0.9421	0.0389	0.7520
Persistence	0.5178	0.9006	0.1864	0.6583

and equivalently close agreement for LightGBM and persistence. This is a second independent training run, ten days after the first, on a maintained codebase that has continued to change elsewhere in the interim (§VI); the reproduction is not a formality.

a) Against persistence the picture is mixed; against LightGBM it is not: The U-Net clears persistence on PR-AUC (gap 0.026) and on Brier by a wide margin (0.0577 against 0.1864), and trails it marginally on ROC-AUC and TSS (by 0.004 and 0.004 respectively) — a mixed result, not a clean win. The persistence baseline here is not a bare carried-forward label but a sigmoid-squashed standardised chlorophyll anomaly, a real, continuous, ranked score; its comparatively poor Brier alongside a competitive ROC-AUC is consistent with that score being reasonably well ordered but poorly calibrated to the 7.59% test-period base rate, not with it being a degenerate baseline that is trivially easy to beat. Against the reloaded LightGBM model the comparison is unambiguous: the U-Net trails on all four metrics simultaneously (PR-AUC gap -0.124), with no metric on which it is competitive.

b) The reloaded baseline agrees with its own published number, which is what makes the comparison trustworthy: The reloaded LightGBM's PR-AUC, 0.6680, sits close to hab_early_warning/readme.md's documented 0.661 (persistence 0.511) for the same model, though not identical — that number's row set passed a different multi-horizon coverage filter, while this run's is t+3-only. The closeness is itself a check: the comparison pipeline is measuring the actual shipped model on real held-out water, not something subtly different that happens to run without error.

c) The exported artifact is consistent with the region it claims to cover: A standalone forecast export (exports/hab_bloom_forecast_spatial.nc, not wired into the platform's serving path) was re-opened for this paper and its finite-value fraction measured directly at 62.4%, matching the Arabian-Sea grid's independently measured real-ocean-cell fraction of approximately 63% — land is masked to NaN correctly, not merely by omission.

V. Discussion

Two explanations for a negative result are worth distinguishing, because they license different next steps. The first is that the implementation is wrong: a bug, a leak, a mismatched row set. The matched-row, reloaded- baseline, twice-reproduced design in §III–IV is built specifically to rule this out, and the close agreement with the tabular model's own published number is the check that it worked. The second is that spatial context, added this way, genuinely does not carry enough additional signal for this task to be worth a harder-to-train model — and that is the more likely reading here, given the U-Net still clears persistence by a real margin and is not failing outright, it is simply behind a strong, already-well-tuned tabular alternative.

a) What this narrows, not just what it rules out: The architecture plan this pilot answers to explicitly staged a spatial-only model before a spatial-plus-temporal one for a stated methodological reason: if a full neighbourhood in one frame does not help, stacking several such frames along time is unlikely to rescue the approach, since temporal recurrence adds a different kind of information (persistence and trend) that a sufficiently strong tabular model can already partly capture through lagged features. A negative U-Net result is therefore a specific, informative data point against reaching for ConvLSTM [2] next by default, not an argument that no spatial architecture could ever help this problem.

VI. Limitations

This is one region (Arabian Sea), one horizon (t+3), one channel set, and one architecture family (a plain U-Net, not a wider search over depth, channel width, or loss function). The codebase continues to change — this paper's own re-verification on 2026-09-09 exists precisely because a repository that trains and retrains models should not be trusted to reproduce a result from prose alone, a discipline this same research programme applies to its own other papers [5]. The exported artifact is explicitly a standalone, unwired output: it is not served to any product surface, and this paper makes no claim that it should be, absent a future spatial attempt that actually beats the point model on held-out skill. Boyce-index spatial calibration, computed for the platform's habitat- suitability models elsewhere, was not populated for this classification comparison and is not reported here rather than reported as a placeholder.

VII. Conclusion

A single-frame U-Net given full spatial context for t+3 harmful-algal-bloom forecasting beats a naive persistence baseline but loses clearly to the existing per-cell LightGBM model on 575,064 identical held-out rows,

reproduced live to three to four decimal places ten days after the original measurement. The result is a real, informative negative: it does not show spatial structure is irrelevant to blooms, only that this way of giving a model access to it does not beat a well-tuned tabular alternative on this task, and it specifically argues against reaching for a spatiotemporal architecture next, since the ordering of this agenda item was designed to make exactly that inference available.

Reproducibility

Every macro in shared/generated/macros.tex names its source and evidence grade.

Pipeline. From machine_learning/: PYTHONPATH=. .venv/bin/python -m hab_early_warning.src.spatial_train builds the dense-cube dataset (hab_early_warning/src/spatial_dataset.py), trains the U-Net (spatial_model.py), and scores it against the reloaded LightGBM artifact and persistence via a subprocess (spatial_lightgbm_subprocess.py) to avoid the torch/joblib segfault described in §III.

Artifacts re-opened directly for this paper. machine_learning/models/hab_early_warning_spatial.pt and machine_learning/exports/hab_bloom_forecast_spatial.nc.

Every number was verified against the file or live re-run producing it as of 2026-09-09.

References

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